Simulation of 2D Collision: A Force-Based Physics Engine

3rd Quarter Physics Group Project - Group 5

Physics Simulation Project

St John Baptist De La Salle Catholic School
Grade 11 A

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Introduction

- **Objective:** to simulate a two–dimensional elastic collision between two pucks using Python and VPython.
- **Method:** Treat collisions as microscopic interactions between internal stiff springs.
- **Tools:** Developed using Python (VPython) and numerical integration.

"Building a physics engine involves simulating the fundamental mechanics of the physical world rather than just the visual result."

The Physics Framework: Hooke's Law

Instead of treating pucks as perfectly rigid, we simulate them with internal springs.

Force Generation

When two pucks overlap, a restorative force (F) is generated:

$$F = k \cdot \text{overlap}$$

- k: Stiffness constant (The "Goldilocks" value).
- **Overlap**: The distance the objects have penetrated each other.

Motion Dynamics: Newton's Second Law

Once the restorative force is calculated, we determine how the objects move.

$$a = \frac{F}{m} \tag{1}$$

- The mass (m) of each puck dictates its resistance to acceleration.
- This allows for collisions between objects of different sizes and masses to feel realistic.

Numerical Integration: The Euler Method

Computers process time in discrete values (dt). We update the system every frame:

1. **Detection:** Is $d < (r_1 + r_2)$?
2. **Overlap:** Overlap $= (r_1 + r_2) - d$
3. **Update Velocity:** $v_{new} = v_{old} + (a \cdot dt)$
4. **Update Position:** $x_{new} = x_{old} + (v_{new} \cdot dt)$

Note: This loop replaces standard algebraic kinematic formulas to allow for dynamic force changes.

Vector Implementation

To resolve the collision in 2D space, we calculate components:

Spatial Calculations:

- $dx = x_2 - x_1$
- $dy = y_2 - y_1$
- $d = \sqrt{dx^2 + dy^2}$

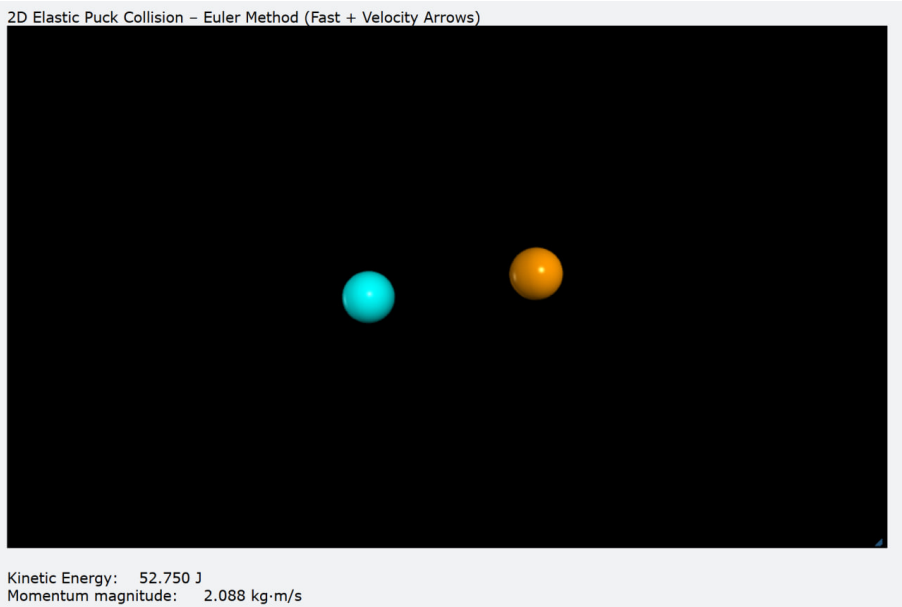
Force Vectors:

- $n_x = \frac{dx}{d}, n_y = \frac{dy}{d}$
- $F_x = F_{mag} \cdot n_x$
- $F_y = F_{mag} \cdot n_y$

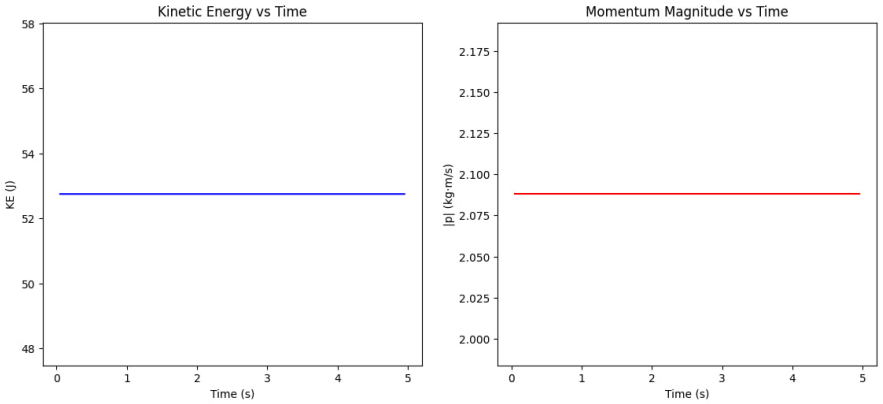
Challenges and Debugging

- **Numerical Instability:** If k is too high or dt is too large, the system gains infinite energy (it "explodes").
- **Energy Drift:** The Euler Method is an approximation; we must monitor for conservation of energy.
- **The Goldilocks Zone:** Finding the perfect balance between stiffness and time-step stability.

Simulation Results



The simulation



The graph showing conservation

- Successfully demonstrated elastic collisions.
- Verified conservation of momentum and kinetic energy.

Conclusion

- Transitioned from static geometry to a dynamic Newtonian system.
- The force-based approach provides a more robust foundation for complex simulations.
- **Open Source:**

github.com/BOSSNATY/2D-Elastic-Collision-Engine

Thank you for your attention!

Questions?